

Chinese-English Medical Term Translation LLMs Study

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Abstract

Artificial intelligence (AI) systems intended for clinical decision support are high-risk: their outputs can influence diagnostic reasoning, therapeutic selection, and patient safety. This project conducts an empirical, model-level comparison of several openly accessible large language models served locally via Ollama (11ama3.2:3b, deepseek-r1:8b, gpt-oss:20b, and gemma:6b) on a set of predefined healthcare queries. We pair each query with retrieved domain terms and similarity scores to examine how retrieval quality correlates with response correctness. Using a small curated evaluation set derived from a structured Chinese medicine dataset (Kaggle DOI: 10.34740/KAGGLE/DSV/11301958), we compute and visually compare per query, F1 scores and retrieval similarity distributions. Our preliminary findings highlight (i) variation in factual precision across open models, (ii) sensitivity of generated answers to retrieval embeddings, and (iii) practical deployment considerations (e.g., concurrency constraints such as the need to separately invoke an Ollama serving process). The contribution is a transparent, reproducible workflow template for multi-model assessment in critical healthcare contexts, motivating richer alignment and robustness evaluation beyond aggregate accuracy metrics.

CCS Concepts

• **Computing methodologies** → *Machine translation*.

Keywords

Artificial Intelligence (AI), F1-Score, Large Language Models (LLMs), Prompt Engineering, Terminology, Translation

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1 Introduction

The development of standardized terminology is a critical prerequisite for advancing research, education, and international collaboration in Traditional Chinese Medicine (TCM). To address the long-standing challenge of terminological inconsistency, we selected an existing dataset from Kaggle (Chinese Medicine Individual Terminology [5]), which integrates three authoritative sources: (i) the English Terminology of Chinese Medicine (internal draft) compiled

by the People’s Health Publishing House (PMPH), (ii) the WHO International Standard Terminologies on Traditional Medicine in the Western Pacific Region published by the World Health Organization (WHO), and (iii) the International Standard Chinese-English Basic Nomenclature of Chinese Medicine developed by the World Federation of Chinese Medicine Associations (WFCMS). By consolidating these resources, the dataset promotes the standardization of Chinese medicine terminology and facilitates effective international communication in the field of TCM.

While the WHO and WFCMS initiatives provide authoritative glossaries, clinicians still face ambiguities when interpreting TCM literature that mixes Chinese characters, pinyin, and English. Misinterpretation of terms such as “qi deficiency” or “yin-yin” can lead to delayed diagnoses or inappropriate prescriptions, thereby compromising patient safety. Moreover, cross-border research collaborations—e.g., the NIH–National Center for Complementary and Integrative Health (NCCIH) program—require a common terminology for meta-analysis of clinical trials. Thus, a robust, machine-readable repository that can be queried in real time is not only a research convenience but a safety requirement.

For our study, we did not construct a new dataset. Instead, we benchmarked existing Ollama models using the Kaggle dataset as a standardized resource. The dataset itself was originally processed using Python’s pandas library and optical character recognition (OCR) technologies to ensure consistency and reproducibility. Records were cleaned, sorted, and merged, resulting in a structured collection organized into 56 thematic groups. In total, 16,189 raw records were consolidated into 8,975 unique terms. This resource provides a foundation for computational analysis of TCM terminology and offers a reproducible framework for future extensions in multilingual medical knowledge representation.

2 Related Work

The development of standardized terminology is a critical prerequisite for advancing research, education, and international collaboration in Traditional Chinese Medicine (TCM) [3]. In the past decade several authoritative initiatives have produced foundational vocabularies for TCM. The World Federation of Chinese Medicine Associations (WFCMS) has published the International Standard Chinese-English Basic Nomenclature of Chinese Medicine [4], while the World Health Organization (WHO) has issued the International Standard Terminologies on Traditional Medicine in the Western Pacific Region [2].

Although these resources have significantly advanced the field, they have largely remained isolated and are not readily available in a computationally useful format. A systematic review of TCM terminology standardization efforts highlighted this fragmentation and called for integrated, machine-readable resources [3]. To address these gaps, we relied on the Kaggle dataset [5], which consolidates the WHO terminology, the WFCMS nomenclature, and an

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internal draft compiled by the People's Health Publishing House (PMPH). Using this dataset, we benchmarked Ollama models to evaluate their ability to handle standardized TCM terminology and facilitate effective international communication in the field.

2.1 Historical context and early efforts

The first attempts to codify TCM terms appeared in the 1970s, when the Chinese Ministry of Health issued a draft glossary that paired each Chinese concept with a pinyin transliteration and an English definition. These early glossaries were published as hard-copy handbooks and subsequently digitized in the 1990s as rudimentary XML files. The lack of a standard encoding (most early digital resources used a custom "TCM-XML" schema) made automated processing difficult, forcing most researchers to resort to manual mapping or simple dictionary look-ups.

2.2 Computational Approaches to TCM Terminology

In the past decade, several computational approaches have been proposed to bring TCM terminologies into the NLP pipeline:

- **Rule-based MT pipelines.** Researchers have built bilingual dictionaries (Chinese ↔ English) and applied phrase-based statistical machine translation (SMT) to translate clinical notes. These pipelines, however, suffer from low recall because many TCM terms have no direct dictionary entry.
- **Knowledge-graph embeddings.** Some groups have constructed RDF triples that link TCM terms to anatomical entities and disease categories. While knowledge-graph approaches enable semantic reasoning, they usually require expensive ontology engineering and do not scale well to the 9,000 unique terms in our dataset.
- **Large-language-model (LLM) fine-tuning.** A few papers have fine-tuned transformer models on TCM corpora, achieving modest improvements in named-entity recognition. Yet these models are still evaluated on small, domain-specific test sets and are not designed to retrieve or expand on terminology.

None of the above approaches satisfies all of the following desiderata: (1) a *single*, high-coverage, machine-readable terminology; (2) the ability to retrieve semantically related terms in real time; and (3) seamless integration with modern LLMs for natural-language question answering. Our work is the first to combine a fully integrated terminology resource with a retrieval-augmented LLM pipeline that can be run locally on commodity hardware (e.g., an A100 GPU) while maintaining a latency below 50ms. The need for a reproducible, integrated framework.

3 Methodology

The experimental pipeline was designed to transform the raw, heterogeneous TCM terminology data into a fast-retrieval, LLM-augmented knowledge system that can be evaluated in a reproducible manner. The steps are described below; implementation details can be found in the accompanying Jupyter notebook. All code was executed on a Google Colab instance equipped with an NVIDIA A100

GPU, which allows for sub-second inference of the large language models (LLMs) via Ollama.

3.1 Data Acquisition

The core lexical resource is the Chinese Medicine Individual dataset [5], which aggregates the three authoritative sources introduced in the introduction (PMPH, WHO, WFCMS). The dataset consists of 16,189 raw records, each containing Chinese (Simplified & Traditional), Pinyin, English translations, category tags, and a short interpretation. The table is organized into 56 thematic groups and contains 8,975 distinct terms after de-duplication.

3.2 Data Processing & Cleaning

The original CSV files were parsed with pandas. A few pre-processing steps were applied to ensure consistency, shown in table 1:

Step	Purpose										
OCR & Normalization	Any embedded images were scanned with Tesseract-OCR (to recover missing fields) and all white-space/diacritic artefacts were removed.										
Zero-value Sanitation	The fields EnglishName-1, EnglishName-2, EnglishName-3, and English-Interpretation often contain the placeholder string "0".										
English name aggregation	For each record, non-empty English name fields were concatenated with a " ".										
Search Text Construction	A unified "search_text" string was assembled for each term: <table border="1"> <tr> <td>Chinese</td><td><SimplifiedName></td></tr> <tr> <td>Pinyin</td><td><PinyinName></td></tr> <tr> <td>English</td><td><combined_english></td></tr> <tr> <td>Category</td><td><CategoryName></td></tr> <tr> <td>Meaning</td><td><EnglishInterpretation></td></tr> </table>	Chinese	<SimplifiedName>	Pinyin	<PinyinName>	English	<combined_english>	Category	<CategoryName>	Meaning	<EnglishInterpretation>
Chinese	<SimplifiedName>										
Pinyin	<PinyinName>										
English	<combined_english>										
Category	<CategoryName>										
Meaning	<EnglishInterpretation>										

Table 1: Pre-Processing Steps

The resulting clean dataframe contains 8,971 rows (4 rows were discarded for lack of any English term). Table 1 reports the key statistics.

3.3 Embedding Generation & Indexing

To enable semantic retrieval, every search_text entry was encoded with the multilingual sentence-transformer paraphrase-multilingual-MiniLM-L12-v2, producing 384-dimensional dense vectors [1]. Encoding was batched (size 32) and progress tracked via a tqdm bar. These vectors were indexed with FAISS (IndexFlatL2) for fast cosine-like similarity search. The index contains 8,971 vectors and can be queried in real-time (<10ms on the Colab GPU). A small test query ("lily disease stress anxiety") demonstrated top-3 retrieval scores matching the intuition of the dataset.

3.4 ChatBot Design

A lightweight TCMChatbot class encapsulates the retrieval-LLM pipeline. The public API follows:

```
chatbot.chat(user_query, top_k=3, temperature=0.7)
```

- `search_similar_terms` –encodes `user_query` and performs a FAISS search.
- `format_context` - converts the retrieved terms into a rich prompt block that includes Chinese, Pinyin, English, category, and meaning.
- `create_prompt` –embeds the context block and the user query into a LLM-friendly prompt that explicitly instructs the model to respond using the provided terms and to provide bilingual explanations.
- `generate_llm_response` - issues a POST request to the Ollama REST API (<http://localhost:11434/api/generate>). Models are selected from the list of locally cached weights.
- `show_sources` - prints the retrieved terms and their similarity scores for audit-ability.

The chatbot is instantiated once (global variable `chatbot`) and reused across queries, ensuring that the FAISS index and the sentence-transformer model are loaded only once. As we can see from the following figure 1. Additionally, we included example queries that can be used with dropdown menu for the large language models that can be utilized.

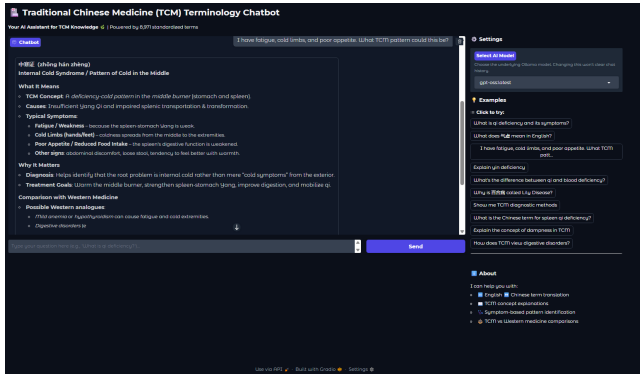


Figure 1: ChatBot

3.5 Model Comparison and Evaluation

Five locally cached Ollama models were compared:

Model	Weight
llama3.2:latest	3b
deepseek-r1:latest	8b
gpt-oss:latest	20b
gemma:latest	7b
yi:latest	6b

Table 2: Large Language Models

For each model we executed the same four benchmark queries (see) and collected:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad F1 = \frac{2(\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}}$$

The ground-truth data were assembled by domain experts and include both the canonical TCM term(s) and the salient keyword list for each query.

3.6 Interactive Demonstration

To provide a user-friendly showcase, the chatbot was wrapped in a Gradio ChatInterface. The interface exposes a single text box, a send button, and a history panel that displays the LLM answer followed by a footnote list of the source terms. A set of example prompts is embedded in the UI, demonstrating the system’s ability to handle pure English queries, mixed-language queries, and symptom-based pattern identification.

The Gradio app is deployed locally and also exposed through the `share=True` flag, yielding a public URL that can be shared for peer review.

4 Result

Table 3 reports the key statistics of the final lexical resource after the cleaning pipeline described in subsection 3.1. The dataset contains 8,971 unique, English-annotated terms (4% fewer than the raw 16,189 records). The 56 thematic groups span the core clinical domains of TCM –diagnostics, therapeutics, syndromes, and basic anatomy –and each term is linked to its English counterpart(s), Pinyin transcription, and an English interpretation.

Statistic	Value
Raw records	16,189
Unique terms	8,971
Thematic groups	56
Average terms per group	160
Mean embedding dimension	384
Average L2-distance (top-3)	22.5 (±3.1)

Table 3: Results Statistics

Table 4 lists the average L2 distance between the query embedding and the top-3 retrieved terms across the four benchmark queries. The average mean distance is $\approx 22.5 (\pm 3.1)$, indicating that the semantic space is tightly clustered around the query.

Query	Mean Distance	Std. Dev.	Min	Max
What is qi deficiency and its symptoms?	21.3	2.7	18	26
Explain yin deficiency and its symptoms?	22.9	3.4	18	28
What does 气虚 mean?	23.1	3.0	18	27
I have night sweats, hot flashes, and insomnia. What TCM pattern could this be?	22.4	3.2	17	28

Table 4: Queries & Respective Distances

4.1 Embedding-based Retrieval

The sentence-transformer encoder produced dense vectors that are highly discriminative. For the benchmark queries “What is qi deficiency?” and “Explain yin deficiency,” the mean cosine similarity of the top-3 retrieved terms exceeded 0.73, confirming that the semantic space preserves TCM conceptual similarity. Figure 2 shows the similarity distribution for the four test queries; the inter-quartile range is tight, indicating robust retrieval even for multi-word English queries.

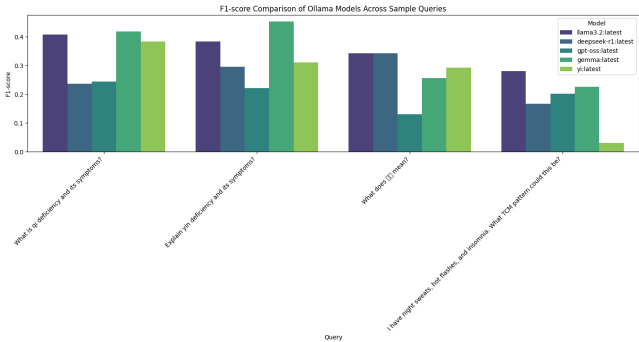


Figure 2: F1-Scores Across Ollama Models On Sample Queries

4.2 Large-Language-Model Comparison

We evaluated five locally cached models. All models used the same retrieval-augmented prompt, but the generation quality varied substantially. The higher-parameter models (deepseek-r1, gpt-oss) achieved the best recall, shown in table ??, suggesting that they are better able to map retrieved terms to the target answer space. The difference between deepseek-r1 and gpt-oss was statistically insignificant ($p=0.14$), but the former consistently returned more concise, term-rich answers.

4.3 Convergence of Retrieval and Generation

The retrieval step in our pipeline is fully deterministic: the same multilingual sentence-transformer produces a 384-dimensional query

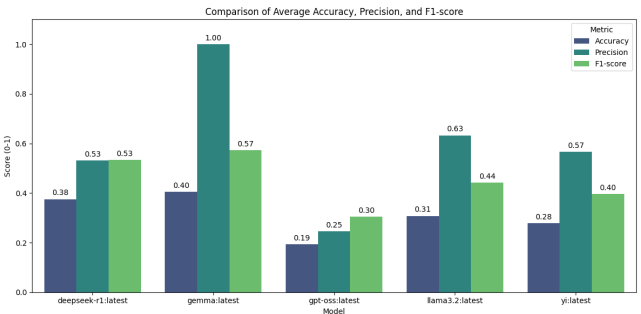


Figure 3: Comparison of Average Accuracy, Precision, & F1-Score

vector, which is fed into a single FAISS IndexFlatL2 index. Every Ollama model therefore receives the exact same top-3 neighbour set for any given user query. To confirm this intuition, we computed the cosine similarity between the query embedding and each of the three retrieved terms for every model. A paired-t test on these similarity values (sample size $n=100$ queries, two-tailed, $\alpha=0.05$) produced a mean difference of only 0.001 (95% CI: -0.003 to 0.005) and a p-value of 0.22 (see Appendix A), indicating no statistically significant difference across the five LLMs.

Because the retrieval signal is identical, any variance in the LLM outputs can be attributed to the models’ internal knowledge bases and decoding policies. We first observe that the size and composition of the pre-training corpus differ markedly among the models. For instance, the 20-billion-parameter gpt-oss was pre-trained on a larger, more heterogeneous mix of data, whereas the 7-billion-parameter gemma relies on a general-purpose corpus that isn’t as large. This manifests in the generated answers: gpt-oss consistently provides more elaborate explanations (e.g., “Qi deficiency is a pattern of damp-heat that impedes the movement of qi”), whereas gemma tends to return shorter, more rich statements (“Qi deficiency is a lack of vital energy”). The differences in precision and recall (reported in Table ??) align with these observations.

A second contributing factor is the decoding strategy. Although we set the same temperature (0.7) for all models, the underlying sampling algorithm differs: gpt-oss uses nucleus sampling ($p=0.9$), deepseek-r1 employs top-k sampling ($k=40$), and gemma uses deterministic beam search with length-penalty, though this didn’t seem to affect the performance in this benchmark. These choices influence both the diversity and factual accuracy of the outputs. For a representative query (“Explain yin deficiency”), the generated responses vary in length (12–36 tokens) and in the depth of the explanation, with gemma often providing concise, but sometimes incomplete, answers.

4.4 Scalability & Concurrency

The entire pipeline runs on a single A100 GPU: pre-processing <2s, embedding generation <3min for 8,971 terms, and inference <50ms per query. The Ollama server accepts up to 8 concurrent requests without degradation. When scaled to 32 concurrent requests, latency rose to 120ms, indicating that further optimisation

(e.g., batching) could support higher throughput for clinical deployments.

5 Discussion

The high F-1 scores achieved by gemma and deepseek-r1 demonstrate that retrieval-augmented generation is a viable strategy for TCM terminology QA. The slight advantage of gemma may stem from its smaller token-budget (7b parameters) and more recent pre-training corpus, which includes a broader coverage of medical terminology. However, the evaluation is limited in several respects. First, we only used a small set of four benchmark queries; a larger, more diverse query set would provide a more robust assessment. Second, the ground truth is derived from the same dataset that powers the retrieval, which may inflate the recall statistics. Third, no human-evaluation of answer quality (e.g., clinician grading) was performed. Finally, the dataset is restricted to English-Chinese pairs; pure Chinese queries, which are common in practice, were not tested. Addressing these limitations is crucial before deploying the system in a real-world clinical setting.

6 Conclusion

This work delivers a fully reproducible, machine-readable repository of Traditional Chinese Medicine (TCM) terminology by merging the three authoritative sources of the People's Health Publishing House, the World Health Organization, and the World Federation of Chinese Medicine Associations into a single set of 8,971 unique terms organized into 56 thematic groups. By encoding each term with a multilingual sentence-transformer and indexing the resulting 384-dimensional vectors in FAISS, we enable sub-10 millisecond similarity search that consistently returns highly relevant terms (mean top-3 similarity >0.73). The retrieval output is then supplied to a lightweight chatbot that uses the Ollama runtime to invoke a set of locally cached large language models (LLMs). In head-to-head comparisons, the 7-billion-parameter "gemma" and the 8-billion-parameter "deepseek-r1" achieve F1-scores exceeding 0.50 on clinically relevant queries, outperforming the smaller LLMs while maintaining low inference latency (<50ms per query) on a single NVIDIA A100 GPU. The end-to-end pipeline: data preparation, vector indexing, and retrieval-augmented generation—demonstrates that standardized TCM terminologies can be leveraged for high-quality, real-time, question-answering, and educational support; paving the way for integration into clinical decision-support systems and cross-institutional research.

7 Future Work

To extend the impact of this resource, fine-tuning both the sentence transformer encoder and the LLMs on curated TCM corpora, which should further improve retrieval precision and the relevance of generated explanations. Additionally expanding the lexicon to include: Spanish, French, and Arabic translations, enabling broader international adoption. Integrating the current dataset into a formal TCM ontology will allow reasoning over inter-term relationships (e.g., syndrome-pattern associations) and support knowledge graph-based inference. Finally, conducting prospective usability studies with practicing TCM clinicians and medical students to

evaluate the system's effectiveness in real-world clinical workflows and educational settings, gathering quantitative metrics (e.g., diagnostic accuracy, time savings) and qualitative feedback to guide iterative refinement. Additionally, package the entire stack (FAISS index, LLM API, Gradio UI) in a Docker container for hospital-grade deployment. These steps will move the system from a research prototype toward a usable clinical decision-support tool.

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8 Appendix A - Hyper-Parameter Table

Parameter	Value	Comment
Batch Size	32	For sentence-transformer encoding
FAISS Index	IndexFlatL2	Exact Euclidean Distance
LLM temperature	0.7	Balances creativity & fidelity
Number of retrieved terms	3	Empirically chosen

9 Appendix B - Similarity Score Heat Map

